CAN Bus Intrusion Detection System

Model Evaluation Report

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Architecture: DistilBERT (4 layers, 4 heads, dim=256)

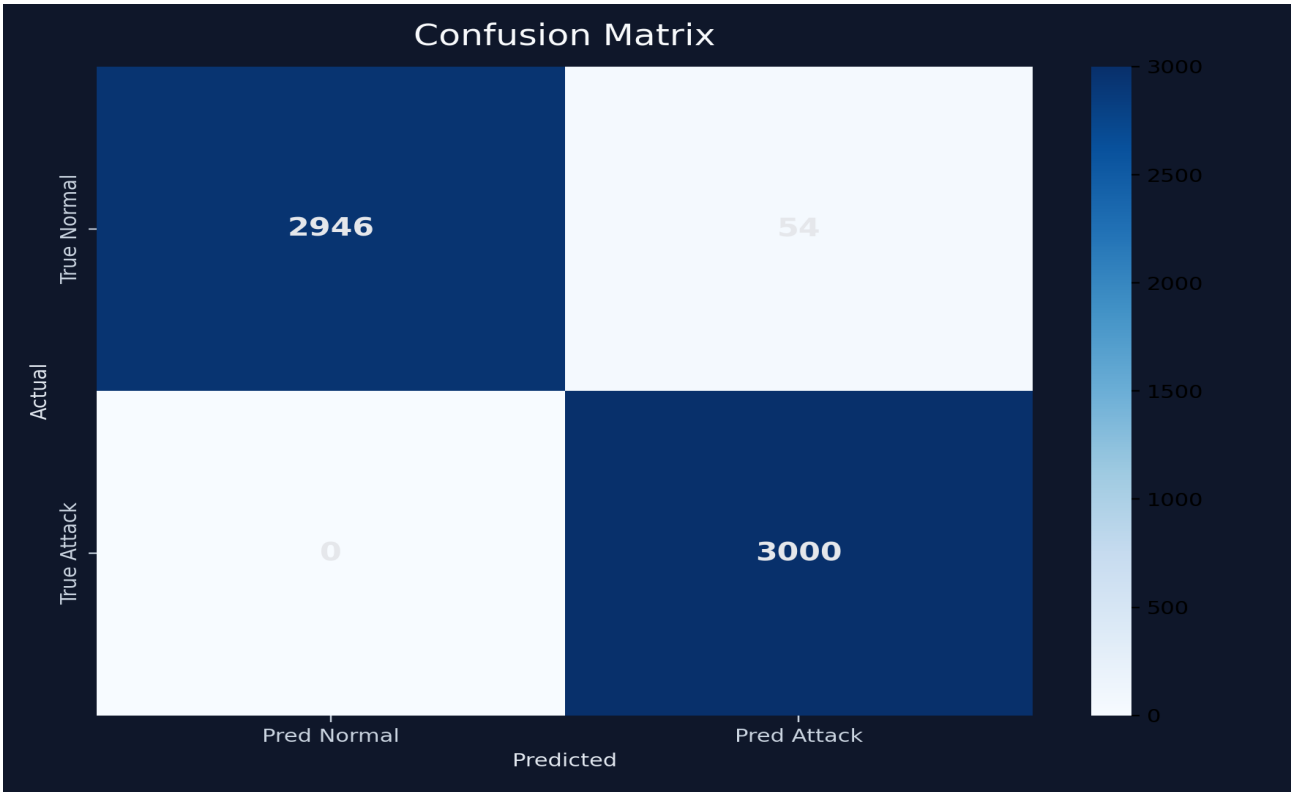
Model Information

Field	Value
Project	CAN Bus Intrusion Detection System
Architecture	DistilBERT (4 layers, 4 heads, dim=256)
Checkpoint Epoch	19
Validation Loss	0.540486
Threshold	3.170000
Threshold Percentile	99
Masking Ratio	0.15
Normal Samples	3000
Attack Samples	3000
Device	cpu

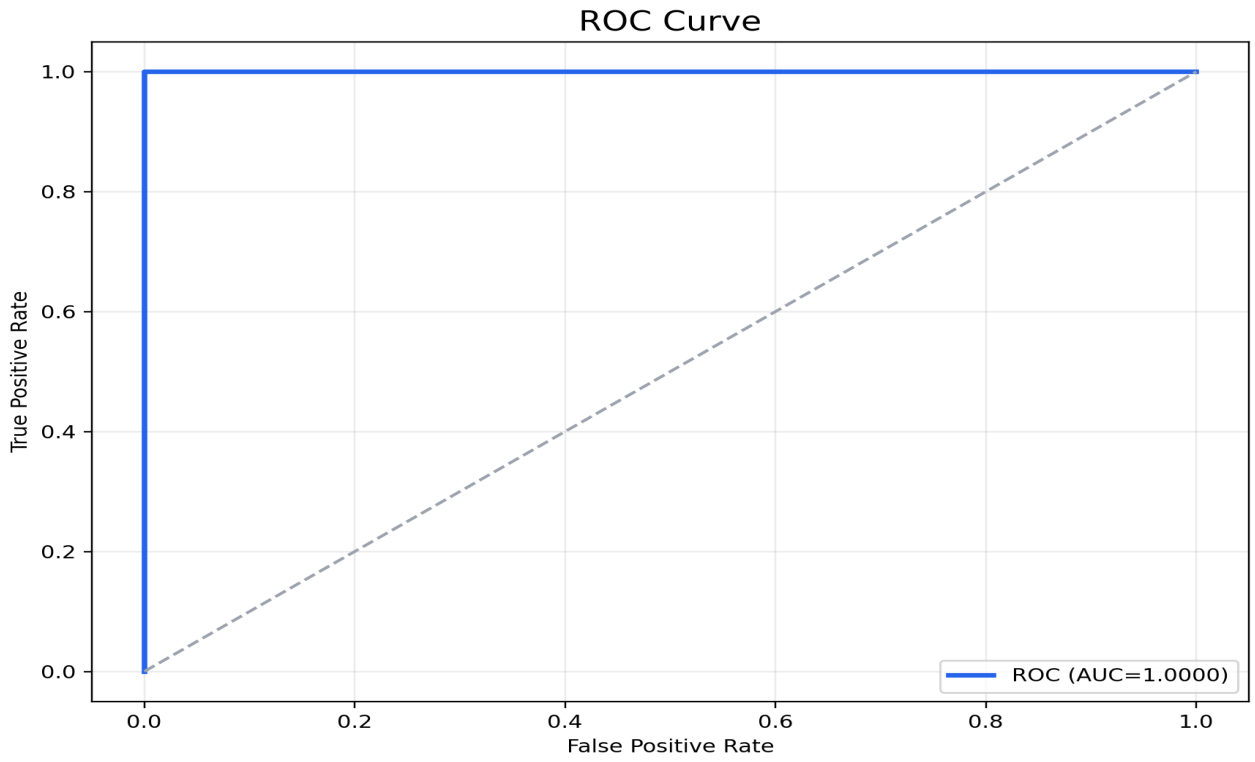
Evaluation Metrics

Metric	Value
Accuracy	0.9910
Precision	0.9823
Recall	1.0000
F1	0.9911
Specificity	0.9820
AUC	1.0000
AP	1.0000

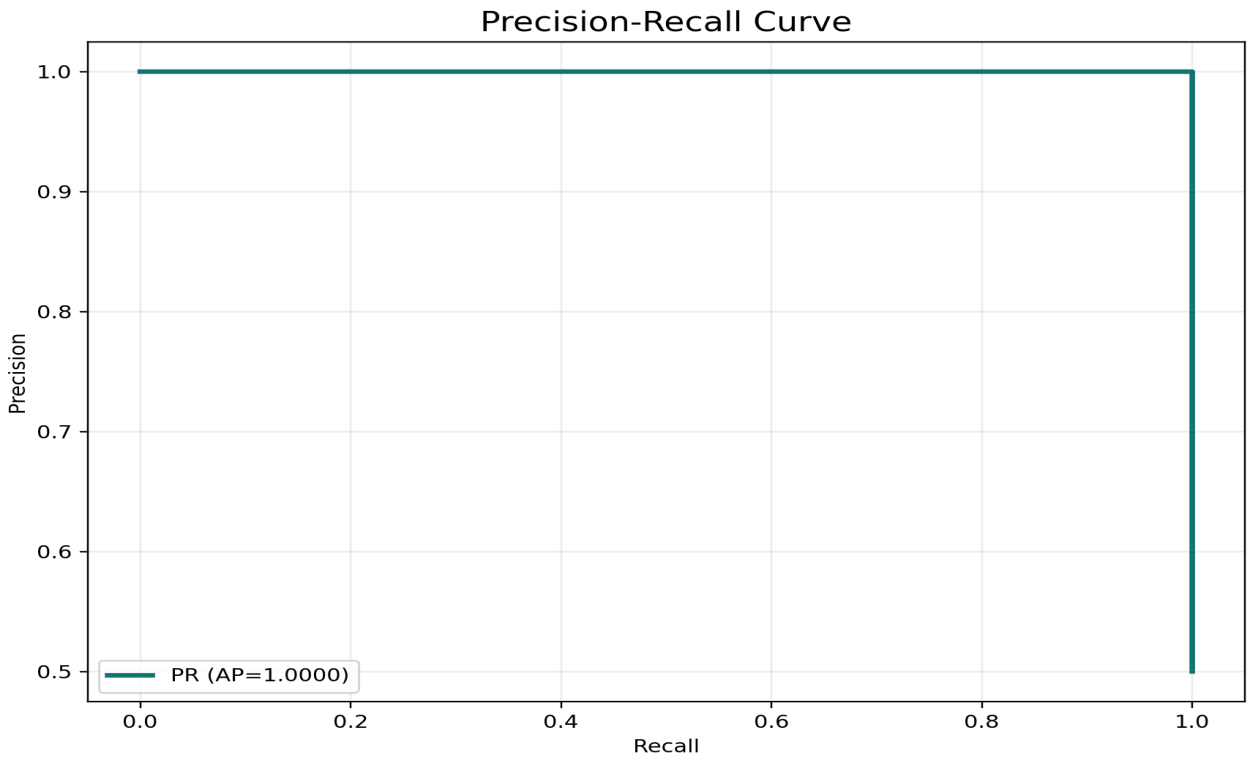
Confusion Matrix



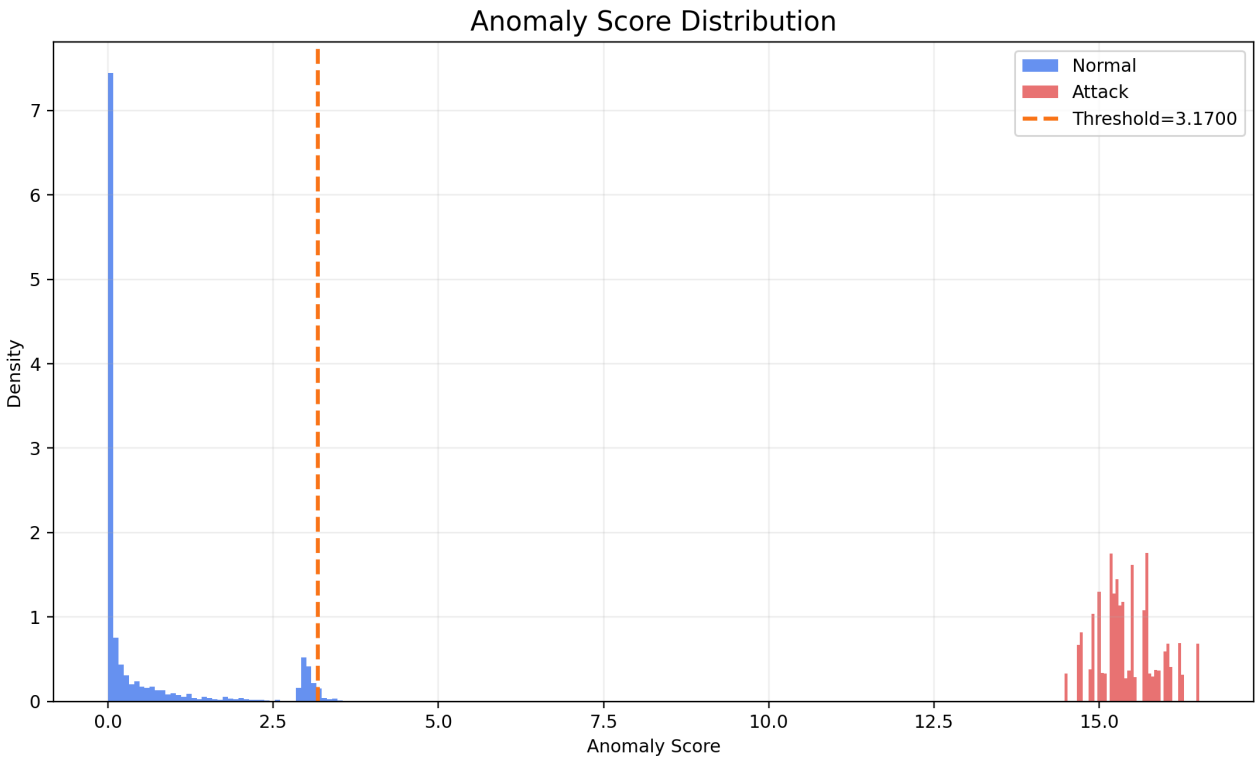
ROC Curve


ROC Curve
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
ROC (AUC=1.0000)

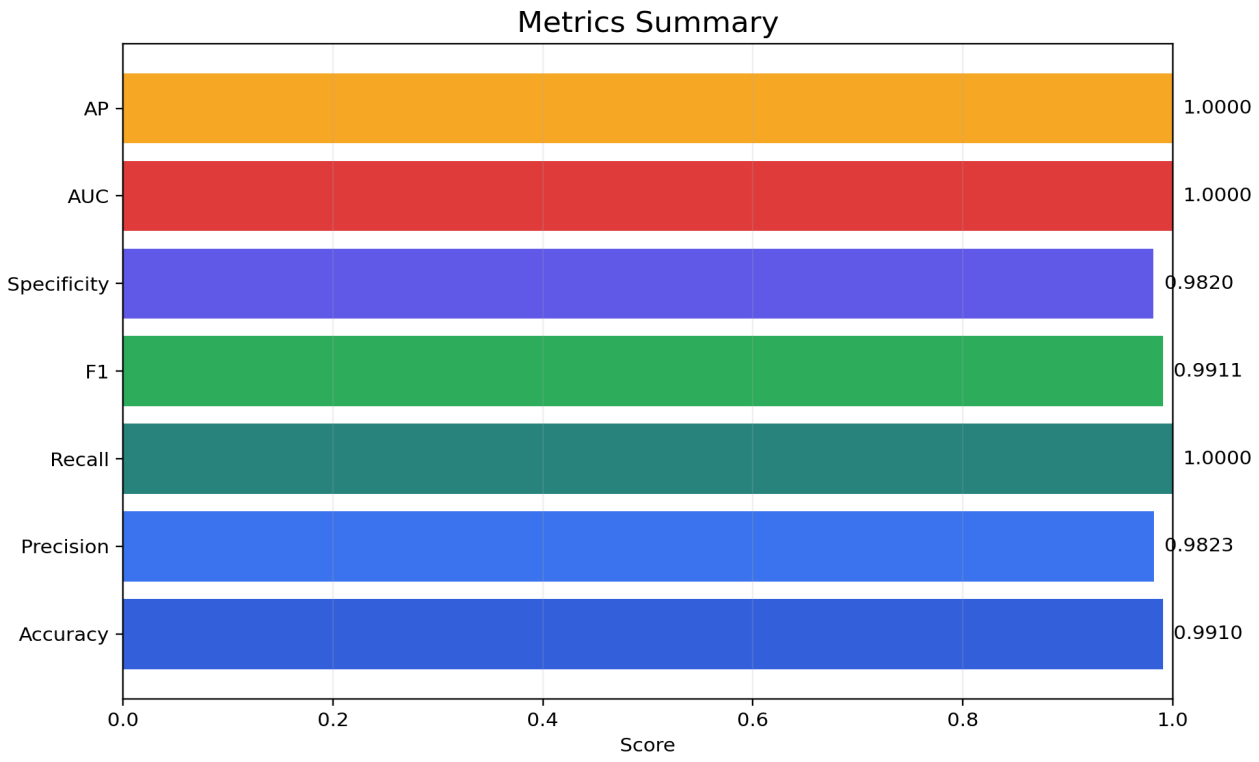
Precision-Recall Curve



Score Distribution



Metrics Summary



Summary Conclusion

The DistilBERT-based CAN IDS demonstrates strong detection performance under the selected threshold (3.1700). The anomaly score distributions indicate separation between normal and attack traffic, while ROC and PR trends confirm ranking quality for attack detection. Threshold calibration and periodic retraining are recommended for long-term deployment stability as vehicle traffic profiles evolve.